José Alfredo Sanabria-Gracia

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Professional Summary

Interdisciplinary Ph.D. researcher trained in inorganic, computational, and bioinorganic chemistry, with hands-on experience spanning synthesis, spectroscopy, electronic structure theory, and data-driven analysis. Proven ability to integrate experimental and computational approaches across chemistry, biology, and materials-related systems. NSF GRFP awardee with a strong interest in collaborative research environments where chemistry supports broader, multidisciplinary solutions to complex scientific and technological problems.

Education

Bachelor of Science: Chemistry **08/2014 – 06/2019**
University of Puerto Rico at Cayey

Post Baccalaureate: Chemistry **07/2021 – 07/2022**
The Ohio State University

Master's Degree Chemistry **08/2022 – 05/2025**
The Ohio State University

Ph.D.: Chemistry **08/2022 – Present**
The Ohio State University

Work History

Analyst in Environmental Quality Laboratories EQ Lab. inc **06/2020 07/2021**
Nancy Villanueva – Bayamón, Puerto Rico

- Perform chemical analysis to determine the presences of 1,1′-dimethyl-4,4′- bipyridilium dichloride salt (DIAQUAT) and 1,1′-ethylene-2,2′-bipyridilium dibromide salt (PARAQUAT) in drinking water samples from Nikkini, Dasani and the water company of Puerto Rico using HPLC with PDA and UV detector. (EPA 549.2)
- Quantification of toxic substances in waste samples with a variety of complex matrixes using GC-MS. (EPA 8260B TCLP)
- Determination of glyphosate in drinking water samples using HPLC with fluorescent detector. (EPA 547)
- Preparation of solutions, instrument calibration and troubleshooting (HPLC and GC), SOP's reading, wet laboratory experience, analytical chemistry experience and data record in laboratory notebooks and data report to my supervisor.

American Academy School Science Teacher **08/2019 – 06/2020**
Ph.D. Viviam Pelayo – Gurabo, Puerto Rico

- Instruct students in the areas of Physics, Biology, Earth Science and Chemistry.
- Ensure the welfare of students during school time.
- Head coordinator of the Science Fair and science activities
- Tutor of students that need reinforcement in Biology, Chemistry, Physics and Math
- Co-coordinator of the organization related to the well care of our planet, Eco-Club.

Academic Experiences

Ph.D. Research in Inorganic Synthesis **01/2023 – Present**

Ph.D. Christine Thomas

- Synthesis of heterobimetallic complexes in a glove box to study properties and reactivity towards small molecule activation like C-H bond functionalization and N_2 fixation.
- Perform DFT calculations to support spectroscopic data and possess a better understanding at the electronic configuration of our systems.
- (NMR, IR, UV-vis) spectroscopy, Gas Chromatography, X-ray Crystallography and Cyclic Voltammetry studies
- Poster presentation at Ohio Inorganic Weekend where second place was awarded.

Ph.D. Research in Physical Bioinorganic **12/2021 – 12/2022**

Ph.D. Hannah Shafaat

- Investigation on a protein-based model of the C-cluster of CODH enzyme to study CO_2 reduction.
- Perform DFT calculations on our [NiFe3S4] protein-based model for insights into mechanism, and electronic structure.
- Electron Paramagnetic Spectroscopy (EPR), Mossbauer Spectroscopy, Gas Chromatography, Uv-vis, Cyclic Voltammetry, X-ray Absorption Spectroscopy (XANES and K-edge at a Synchrotron facility in Sandford) and Resonance Raman studies
- Cell growth and protein purification with Fast Performance Liquid Chromatography (FPLC), SDS PAGE and Buffer Exchange Methods.

B.S. Research in Computational Chemistry **06/2017 - 05/2019**

Ph. D. Juan Santana

- Perform DFT calculations on Lithium-Sulfur battery models to study the anchoring mechanism of Lithium-Polysulfides in the cathode of the battery.
- Use of Python, Gaussian, and ORCA to write input files scripts and set up calculations of energy and spectroscopy parameters.
- Use software like Material Studio and VESTA, GauseView, Gabedit and Avogadro to visualized molecular models and plot data.

B.S. Research in Inorganic Chemistry **06/2017 - 05/2019**

Ph. D. Ramón Hernández

- Synthesis of Coordination Complexes with chemotherapeutic potential using known synthetic paths and Cis-Platin as a starting point.
- Literature research related with Cis-Platin and synthetic paths for coordination complexes.
- Using spectroscopical techniques like NMR and IR for the characterization of reactions products.

B.S. Research in Biochemistry **01/2017 – 03/2017**

Ph. D. Vibha Bansal

- Research with cellulose membranes for the development of an alternative for a physical separation method.
- Conducted activity assays on proteins, protein estimation assays, protein labeling assay and buffer preparation.

Research in Computational Chemistry QUIM 4166 Course **01/2019 - 05/2019**

Ph. D. Dalvin Mendez

- Research in the quantification of simple molecule’s potential energy to predict, the energy of complex molecules.
- Run calculations and developing input file scripts for molecules of interest in programs like python and ORCA.

Research in Analytical Chemistry Capstone Course **08/2018 - 12/2018**
Ph. D. Christian Menendez

- Research in the extraction and quantification of Iron (Fe) from *Spinacea Oleracea* (Spinach) using methods of attaching fluorophores and applying UV-vis and Atomic Absorption Spectroscopy (AAS)
- Preparation of solutions and calibration curves for the quantification as well as running controls to increase confidence in the obtained data.

Research in Organic Chemistry QUIM 3154 Course **01/2016 - 05/2016**
Ph. D. Luz Torres

- Research in the extraction of polyphenols form the husk of Cocoa beans using a Soxhlet system, assisted ultrasound extraction and assisted microwaves extraction methods.
- Solution preparation and waste disposal.

Publications

Lewis, L. C., Sanabria-Gracia, J. A., Lee, Y., Jenkins, A. J., & Shafaat, H. S. (2024). Electronic isomerism in a heterometallic nickel–iron–sulfur cluster models substrate binding and cyanide inhibition of carbon monoxide dehydrogenase. Chemical Science (Royal Society of Chemistry: 2010), 15(16), 5916–5928. https://doi.org/10.1039/d4sc00023d

Title: In Process of writing
PI: Christine Thomas
Status: In process of writing

Awards & Recognitions

Second place Poster Presentation at Ohio Inorganic Weekend Conference (2023). Certificate

NSF-GRFP Awarded in 2023

Special recognition from the Senate of Puerto Rico for the effort and dedication with the education imparted to the students from American Academy during the Covid-19 situation (2019). ***Singed Official Document***

Special recognition from the Stars of the Future class, for your great sense of improvement that serves as an example for others to imitate your dedication and effort by not giving up after 6 surgeries in legs. (**2008)** ***Recognition Plaque***

Special recognition from the Faculty of Sciences in Edmundo Del Valle Cruz school for the dedication and work of excellence in the science fair representing the school at district and regional level every year from first grade to 9th grade. **(2008)** ***Recognition Plaque***

Presentations

Ohio Inorganic Weekend Conference (2025) "Selective C-H Bond Activation in Pyridines Using a Zr/Co Bimetallic Complex Through Metal-Metal Cooperativity" Kent, Ohio.

NOBBACChE Excellence Symposium (2024) "C-H Bond Activation and Functionalization Enabled by a Zr/Co Bimetallic System Through Metal-Metal Cooperativity" Columbus, Ohio

Ohio Inorganic Weekend Conference (2023) “Selective C-H Bond Activation in Pyridines Using a Zr/Co Bimetallic Complex Through Metal-Metal Cooperativity” Bowling Green, Ohio. **(Second Place in Poster Presentation)**

- Invited by the Chemistry Circle ACS Chapter to talk about the Bridge Program in my to university in Puerto Rico. (2021) “The Ohio State University's Bridge Program: My path into research and personal development” Via Zoom.

- SACNAS (2018) “Density Functional Calculations of the Atomic Conformation of Sulfur on Ni(111) Under High Coverage Conditions” San Antonio, Texas.

- QUIM 4999 course UPR-Cayey (2018) “Density Funtional Calculations of the Atomic Conformation of Sulfur on FCC Metal Under High Coverage Conditions” Cayey, Puerto Rico.

Volunteer Experience

Recruitment in Recruitment Week for students in the Thomas research group.

Recruitment in SACNAS 2023 from the Department of Arts and Sciences of Ohio State University

“*Teo-Labs from the Theory to the Experimentation*” Project Ph. D. Wanda Peña & B.S. Teresa Cordero

- **Founder of the program for high school students that needed reinforcement for the transition to an undergraduate degree.**
- Co-coordinator of the laboratory course
- Preparation of the handouts with the materials and procedure
- Preparation of solutions and stations before the experiments
- Disposal of the waste after the experiment
- Supervise the students during the experiment and provide feedback.
- Grade their laboratory reports.

Associations

NOBBChE Ohio State University-student chapter

American Chemical Association-student chapter

SCANAS Ohio State University-student chapter

“*Renacer Navideño*” Festival
Mr. David Ramos – Arroyo, Puerto Rico

- Co-coordinator of the inscriptions for the Marathon “*Abelardo Diaz Alfaro*”
- Supervise for the good care of the runners.
- Announce the winners and special awards on stage.

Family, Careers and Community Lieders of America Ms. Loyda Rodríguez – Arroyo, Puerto Rico (2013 – 2014)

- President of the organization in Arroyo

- Vice-president of the committee in Caguas
- Organize activities to ensure and promote the collaboration between the community.
- Represent the school in the march with Raymond Arrieta
- Organize the celebration of Earth Day

Skills

Data analysis in softwares like Empower, Open Lab, LIMS, Mester Nova, Qsoas, MATLAB, Mercury, Microsoft Excel, Power Point, Word, Igor, Logger Pro and GMP, SOP and WI

Experience with Python, Gaussian, ORCA.

Molecule modeling with VESTA, GaussView, Material Studio, Gabedit, Spartan, WebMO, PyMol, IQmol,

Laboratory techniques and Methods (Wet chemistry): HPLC-PDA, UV-vis and Fluorescence detector, GC-MS, GC-FID, Cyclic Voltammetry, Titration, Decantation, Simple and Fractionated destilation and other separation methods, Extraction methods, Pipetting, Crystallization, TLC, Reflux, Solution preparation, filtration, and synthesis.

Spectroscopy: NMR, IR, MS, Uv-vis, EPR, Mossbauer, XAS (XANES and K-edge), AAS and Raman

Trilingual: Spanish (Native), English (Fluent) and Portuguese (Basic)